

Покрокова інструкція виконання фінального проєкту

Технічні вимоги

- **Інфраструктура:** AWS з використанням Terraform
- **Компоненти:** VPC, EKS, RDS, ECR, Jenkins, Argo CD, Prometheus, Grafana

Етапи виконання

1. Підготовка середовища:

- Ініціалізувати Terraform.
- Перевірити всі необхідні змінні та параметри.

2. Розгортання інфраструктури:

- Виконати команду розгортання:

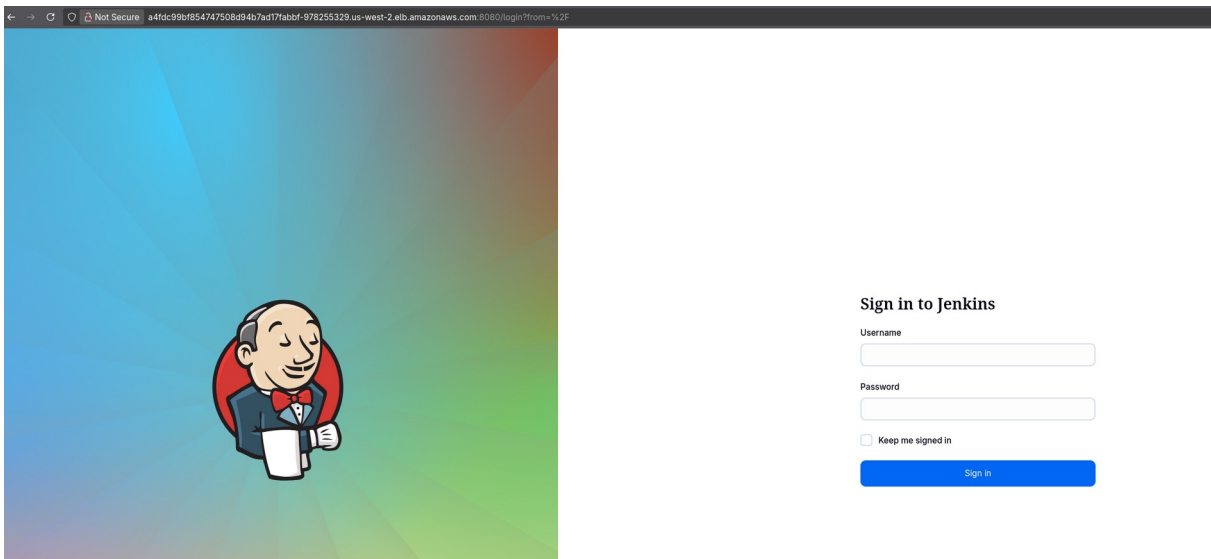
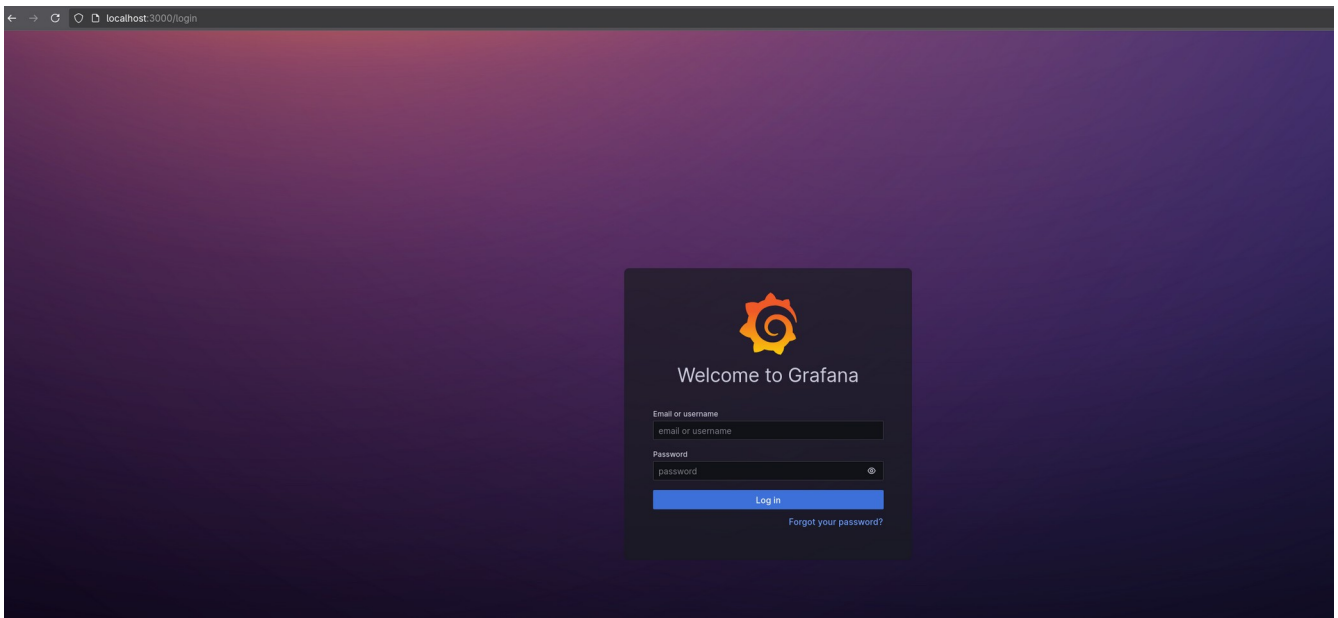
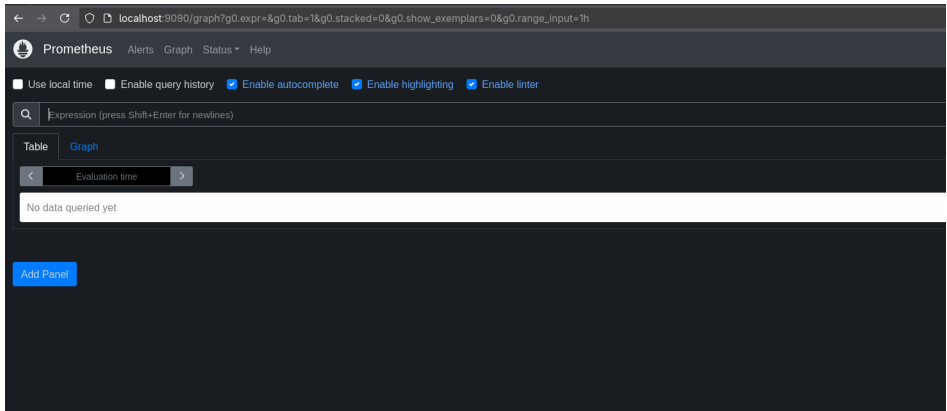
terraform apply

```
... (venv) ~/projects/friendly-bash/Project$ terraform apply "tf.plan"
module.ecr.aws_ecr_repository.this: Creating...
module.eks.aws_iam_role.node_role: Creating...
module.rds.aws_db_parameter_group.this: Creating...
module.eks.aws_iam_role.eks_cluster_role: Creating...
module.vpc.aws_vpc.this: Creating...
module.eks.aws_iam_role.eks_cluster_role: Creation complete after 1s [id=final-eks-eks-cluster-role]
module.eks.aws_iam_role_policy_attachment.eks_cluster_AmazonEKSVPCResourceController: Creating...
module.eks.aws_iam_role_policy_attachment.eks_cluster_AmazonEKSClusterPolicy: Creating...
module.eks.aws_iam_role.node_role: Creation complete after 1s [id=final-eks-node-role]
module.eks.aws_iam_role_policy_attachment.node_AmazonEC2ContainerRegistryReadOnly: Creating...
module.eks.aws_iam_role_policy_attachment.node_AmazonEKSWorkerNodePolicy: Creating...
module.eks.aws_iam_role_policy_attachment.node_AmazonEKS_CNI_Policy: Creating...
module.ecr.aws_ecr_repository.this: Creation complete after 2s [id=final-django]
module.eks.aws_iam_role_policy_attachment.eks_cluster_AmazonEKSClusterPolicy: Creation complete after 1s [id=final-eks-eks-cluster-role/arn:aws:iam::aws:policy/AmazonEKSClusterPolicy]
module.eks.aws_iam_role_policy_attachment.eks_cluster_AmazonEKSVPCResourceController: Creation complete after 1s [id=final-eks-eks-cluster-role/arn:aws:iam::aws:policy/AmazonEKSVPCResourceController]
module.eks.aws_iam_role_policy_attachment.node_AmazonEC2ContainerRegistryReadOnly: Creation complete after 1s [id=final-eks-node-role/arn:aws:iam::aws:policy/AmazonEC2ContainerRegistryReadOnly]
module.eks.aws_iam_role_policy_attachment.node_AmazonEKSWorkerNodePolicy: Creation complete after 1s [id=final-eks-node-role/arn:aws:iam::aws:policy/AmazonEKSWorkerNodePolicy]
module.eks.aws_iam_role_policy_attachment.node_AmazonEKS_CNI_Policy: Creation complete after 1s [id=final-eks-node-role/arn:aws:iam::aws:policy/AmazonEKS_CNI_Policy]
module.vpc.aws_vpc.this: Still creating... [00m10s elapsed]
module.vpc.aws_vpc.this: Creation complete after 14s [id=vpc-0f263027add3b88e3]
module.vpc.aws_internet_gateway.igw: Creating...
module.vpc.aws_route_table.public: Creating...
module.vpc.aws_subnet.public["us-west-2a"]: Creating...
module.vpc.aws_subnet.private["us-west-2a"]: Creating...
module.vpc.aws_subnet.private["us-west-2c"]: Creating...
module.vpc.aws_subnet.private["us-west-2a"]: Creating...
module.vpc.aws_subnet.public["us-west-2b"]: Creating...
module.rds.aws_security_group.this: Creating...
module.eks.aws_security_group.cluster: Creating...
module.vpc.aws_internet_gateway.igw: Creation complete after 1s [id=igw-0c96dc9e5439bb79]
module.vpc.aws_route_table.public: Creation complete after 1s [id=rtb-0e025cdc9c8d936c]
module.vpc.aws_route.default_internet_access: Creating...
module.vpc.aws_subnet.private["us-west-2c"]: Creation complete after 2s [id=subnet-07e1a584614fb21a9]
module.vpc.aws_route.default_internet_access: Creation complete after 2s [id=rtb-0e025cdc9c8d936c/0802B9494]
module.rds.aws_security_group.this: Creation complete after 4s [id=sg-02d750047b948ddde]
module.rds.aws_vpc_security_group_egress_rule.egress_all: Creating...
module.vpc.aws_subnet.private["us-west-2b"]: Creation complete after 4s [id=subnet-0408a5ff9d9db97c54]
module.rds.aws_vpc_security_group_egress_rule.egress_all: Creation complete after 0s [id=sg-09ec87a75d2da3e45]
module.eks.aws_security_group.cluster: Creation complete after 4s [id=sg-002c8d7e1673853d8]
module.vpc.aws_subnet.private["us-west-2a"]: Creation complete after 5s [id=subnet-0a95b02ad3708c12a]
module.rds.aws_db_subnet_group.this: Creating...
module.rds.aws_db_subnet_group.this: Creation complete after 2s [id=rds-subnet-group]
module.vpc.aws_subnet.public["us-west-2c"]: Still creating... [00m10s elapsed]
module.vpc.aws_subnet.public["us-west-2a"]: Still creating... [00m10s elapsed]
module.vpc.aws_subnet.public["us-west-2b"]: Still creating... [00m10s elapsed]
module.vpc.aws_subnet.private["us-west-2c"]: Creation complete after 12s [id=subnet-0f41f3e504c49a1dd]
module.vpc.aws_subnet.private["us-west-2b"]: Creation complete after 15s [id=subnet-0545a1c463ef1a87]
```

- Перевірити стан ресурсів через:

...

```
kubectl get all -n jenkins
kubectl get all -n argocd
kubectl get all -n monitoring
...
```



3. Перевірка доступності:

- Jenkins:

kubectl port-forward svc/jenkins 8080:8080 -n jenkins

```
kyryl@mini:~/IdeaProjects/final-devops-project$ kubectl get all -n jenkins
NAME                READY   STATUS    RESTARTS   AGE
pod/jenkins-0       2/2     Running   0           15h

NAME                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/jenkins     LoadBalancer  172.20.196.129   a4fdc99bf854747508d9ab7ad17fab0f-978255329.us-west-2.elb.amazonaws.com  8080:31011/TCP  15h
service/jenkins-agent ClusterIP      172.20.64.14    <none>           50000/TCP        15h

NAME                READY   AGE
statefulset.apps/jenkins 1/1     15h
kyryl@mini:~/IdeaProjects/final-devops-project$
```

- Argo CD:

kubectl port-forward svc/argocd-server 8081:443 -n argocd

```
^Ckyryl@mini:~/IdeaProjects/final-devops-project$ kubectl port-forward svc/argocd-server 8081:443 -n argocd
Forwarding from 127.0.0.1:8081 -> 8080
Forwarding from [::1]:8081 -> 8080
```

```
kyryl@mini:~/IdeaProjects/final-devops-project$ kubectl get all -n argocd
NAME                                READY   STATUS    RESTARTS   AGE
pod/argocd-application-controller-0 1/1     Running   0           21h
pod/argocd-applicationset-controller-b5c8dbdd-5kzjf 1/1     Running   0           21h
pod/argocd-dex-server-7b6c6d469-w2bch 1/1     Running   0           21h
pod/argocd-notifications-controller-754c4bfbc-4b2q8 1/1     Running   0           21h
pod/argocd-redis-f57676fc7-nstew 1/1     Running   0           21h
pod/argocd-repo-server-5978b98cc9-g8hmx 1/1     Running   0           21h
pod/argocd-server-d784487d-wfabj 1/1     Running   0           87m

NAME                                TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/argocd-applicationset-controller ClusterIP      172.20.202.202   <none>           7000/TCP          21h
service/argocd-dex-server            ClusterIP      172.20.33.45    <none>           5556/TCP, 5557/TCP 21h
service/argocd-redis                 ClusterIP      172.20.68.158   <none>           6379/TCP          21h
service/argocd-repo-server           ClusterIP      172.20.71.246   <none>           8081/TCP          21h
service/argocd-server                 LoadBalancer  172.20.66.27    a21051194c84c4e30ab9548cd6dc0800-1822634096.us-west-2.elb.amazonaws.com  80:32492/TCP, 443:31269/TCP 21h

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/argocd-applicationset-controller 1/1     1             1           21h
deployment.apps/argocd-dex-server 1/1     1             1           21h
deployment.apps/argocd-notifications-controller 1/1     1             1           21h
deployment.apps/argocd-redis 1/1     1             1           21h
deployment.apps/argocd-repo-server 1/1     1             1           21h
deployment.apps/argocd-server 1/1     1             1           21h

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/argocd-applicationset-controller-b5c8dbdd 1           1           1       21h
replicaset.apps/argocd-dex-server-7b6c6d469 1           1           1       21h
replicaset.apps/argocd-notifications-controller-754c4bfbc 1           1           1       21h
replicaset.apps/argocd-redis-f57676fc7 1           1           1       21h
replicaset.apps/argocd-repo-server-5978b98cc9 1           1           1       21h
replicaset.apps/argocd-server-d654dbf44 0           0           0       21h
replicaset.apps/argocd-server-d784487d 1           1           1       87m

NAME                                READY   AGE
statefulset.apps/argocd-application-controller 1/1     21h
kyryl@mini:~/IdeaProjects/final-devops-project$
```

4.

Моніторинг та перевірка метрик:

- Grafana:

kubectl port-forward svc/grafana 3000:80 -n monitoring

```
kyryl@mini:~/IdeaProjects/final-devops-project$ kubectl port-forward svc/prometheus-grafana 3000:80 -n monitoring
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

```
kyryl@mini:~/IdeaProjects/final-devops-project$ kubectl get all -n monitoring

NAME                                READY    STATUS    RESTARTS   AGE
pod/alertmanager-prometheus-kube-prometheus-alertmanager-0  2/2      Running   0           96m
pod/prometheus-grafana-7646d59775-btdrq  3/3      Running   0           96m
pod/prometheus-kube-prometheus-operator-775d58dc6b-lb9c8  1/1      Running   0           96m
pod/prometheus-kube-state-metrics-856b96f64d-r9t8g  1/1      Running   0           96m
pod/prometheus-prometheus-kube-prometheus-prometheus-0  2/2      Running   0           96m
pod/prometheus-prometheus-node-exporter-9mwdc  1/1      Running   0           96m
pod/prometheus-prometheus-node-exporter-vx7k2  1/1      Running   0           96m

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/alertmanager-operated      ClusterIP      None           <none>         9093/TCP,9094/TCP,9094/UDP  96m
service/prometheus-grafana         ClusterIP      172.20.244.57 <none>         80/TCP           96m
service/prometheus-kube-prometheus-alertmanager ClusterIP      172.20.163.25 <none>         9093/TCP,8080/TCP  96m
service/prometheus-kube-prometheus-operator ClusterIP      172.20.128.143 <none>         443/TCP          96m
service/prometheus-kube-prometheus-prometheus ClusterIP      172.20.187.215 <none>         9090/TCP,8080/TCP  96m
service/prometheus-kube-state-metrics ClusterIP      172.20.219.128 <none>         8080/TCP          96m
service/prometheus-operated         ClusterIP      None           <none>         9090/TCP          96m
service/prometheus-prometheus-node-exporter ClusterIP      172.20.142.10  <none>         9100/TCP          96m

NAME                                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE    NODE SELECTOR    AGE
daemonset.apps/prometheus-prometheus-node-exporter  2           2           2           2           2           kubernetes.io/os=linux  96m

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/prometheus-grafana  1/1      1           1           96m
deployment.apps/prometheus-kube-prometheus-operator  1/1      1           1           96m
deployment.apps/prometheus-kube-state-metrics  1/1      1           1           96m

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/prometheus-grafana-7646d59775  1           1           1           96m
replicaset.apps/prometheus-kube-prometheus-operator-775d58dc6b  1           1           1           96m
replicaset.apps/prometheus-kube-state-metrics-856b96f64d  1           1           1           96m

NAME                                READY    AGE
statefulset.apps/alertmanager-prometheus-kube-prometheus-alertmanager  1/1      96m
statefulset.apps/prometheus-prometheus-kube-prometheus-prometheus  1/1      96m
```